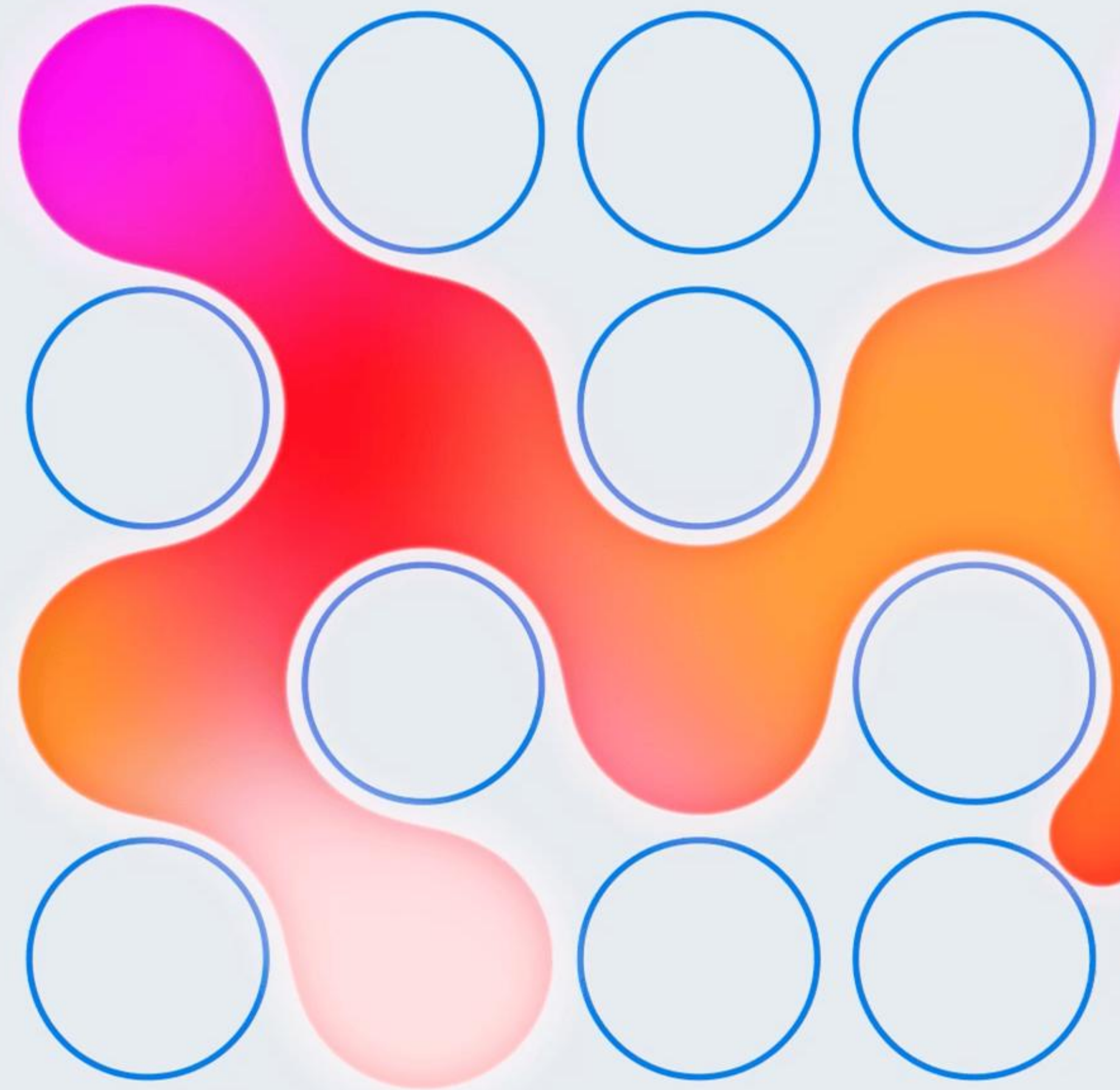




Building agentic systems on AWS

Workshop: Introduction to AI agents



Presenters

Leonardo Murillo

Developer Relations, AWS Marketplace



Thaddeus Worsnop

Principal, Solutions Architect,
Amazon Web Services (AWS)

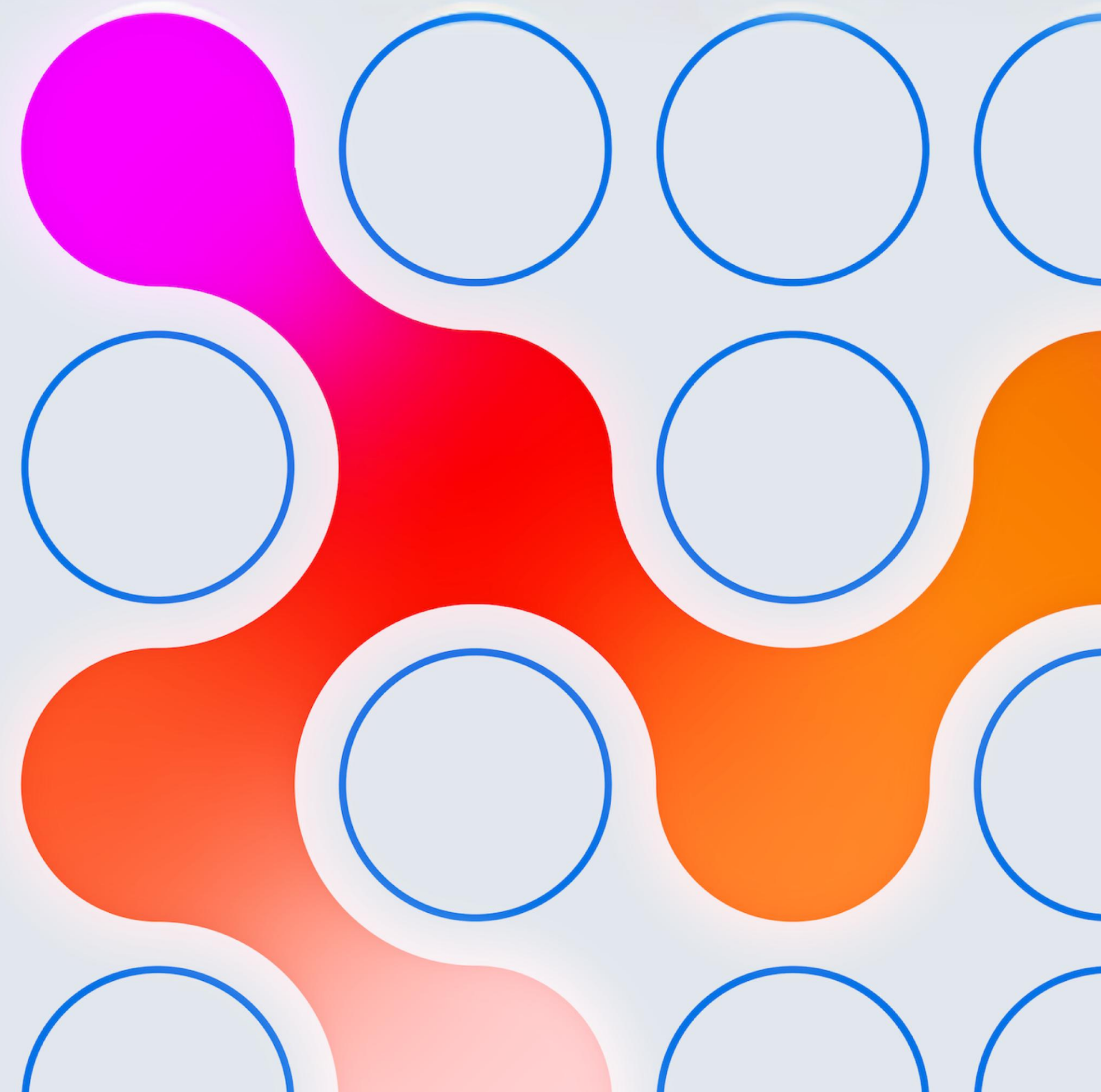


Agenda

- A brief history of AI
- Large language models (LLMs)
- From responses to actions
- Agentic systems architecture
- Enterprise fit and adoption
- Demo
- Next steps

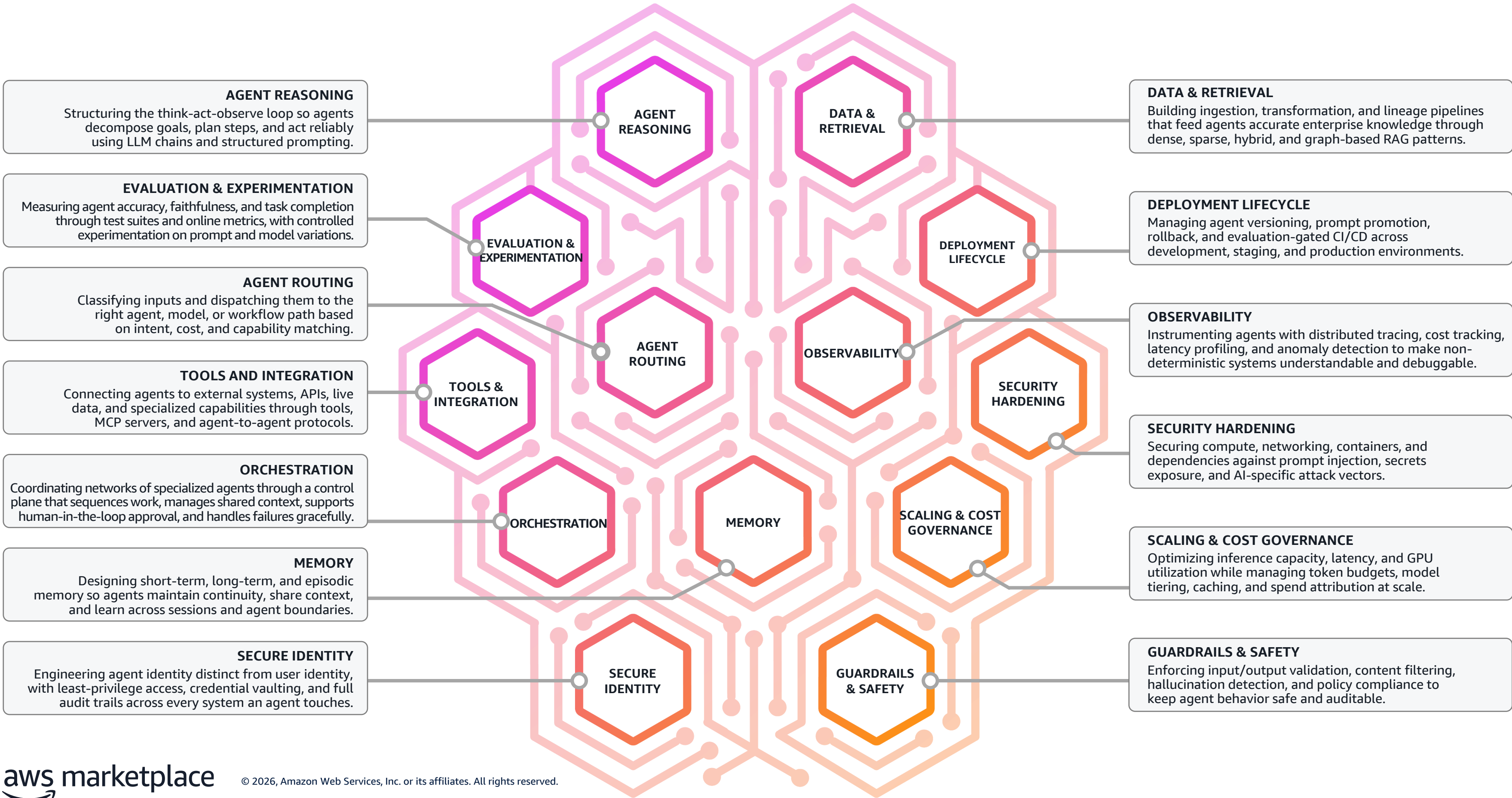


Building agentic systems on AWS



The Agentic AI capability stack

13 capability domains to build production-ready agentic systems



Agentic AI tools and capabilities in AWS Marketplace

AGENT REASONING

Claude

cohere

Hugging Face

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Progress Agentic RAG

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Amazon Bedrock Knowledge Bases

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DEPLOYMENT LIFECYCLE

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INSTANA

logz.io

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AWS X-Ray

AWS Distro for OpenTelemetry

SECURITY HARDENING

snyk

Protect AI

Prompt

tailscale

Amazon GuardDuty

AWS Security Hub

Amazon Inspector

SCALING & COST GOVERNANCE

ClickHouse

NVIDIA AI ENTERPRISE

Modal

cast.ai

baseten

anyscale

hydrolix

StormForge

Amazon EKS

AWS Fargate

Amazon Bedrock AgentCore (Runtime)

GUARDRAILS & SAFETY

Claude

Guardrails AI

Arthur

Amazon Bedrock Guardrails

Amazon Bedrock AgentCore (Policy)

2,800+

Agentic AI tools and capabilities

To help you build, deploy, and operate agentic systems on AWS — available through your AWS account.

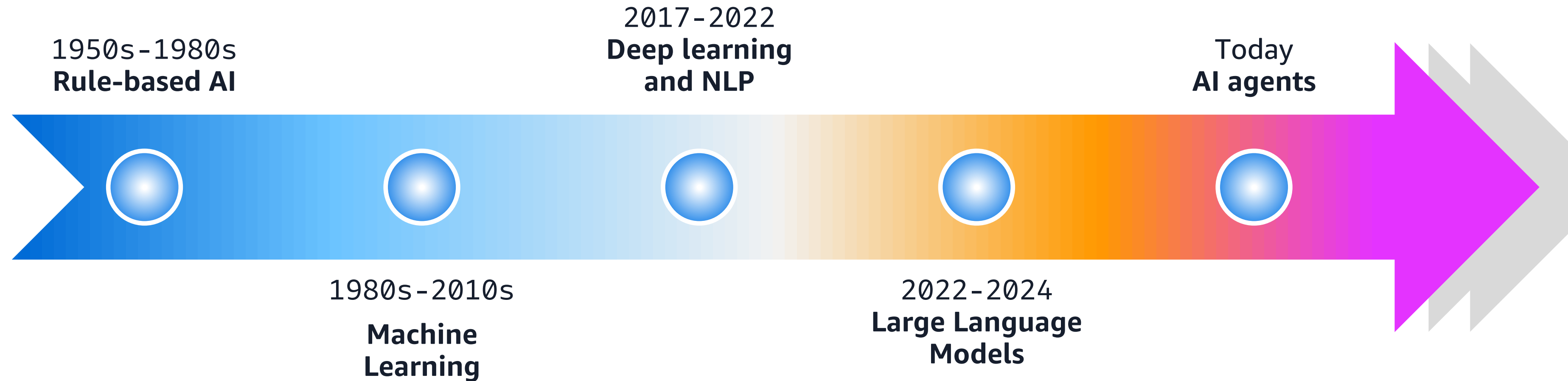
aws marketplace

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A brief history of artificial intelligence

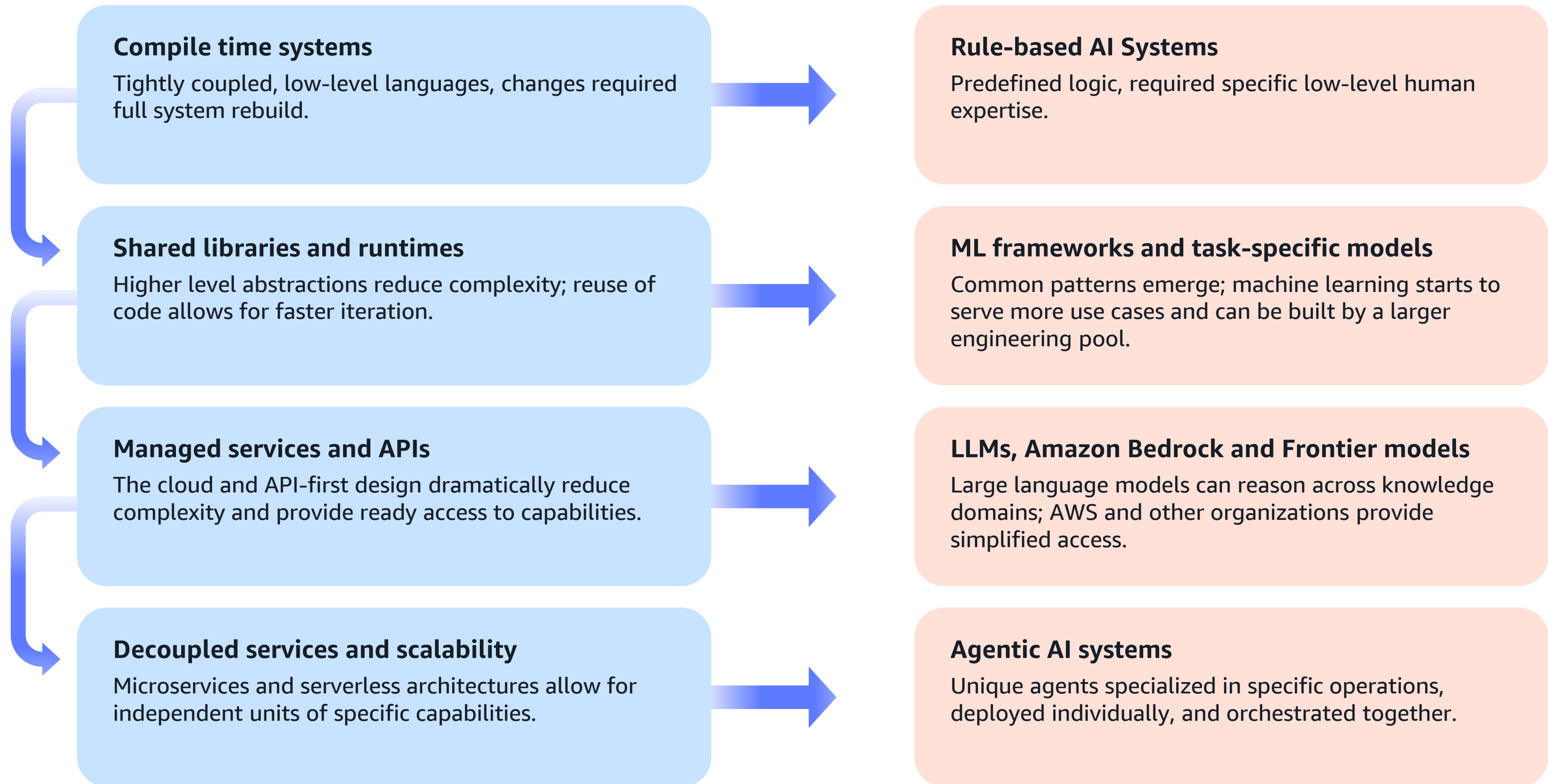
Evolution to agentic systems



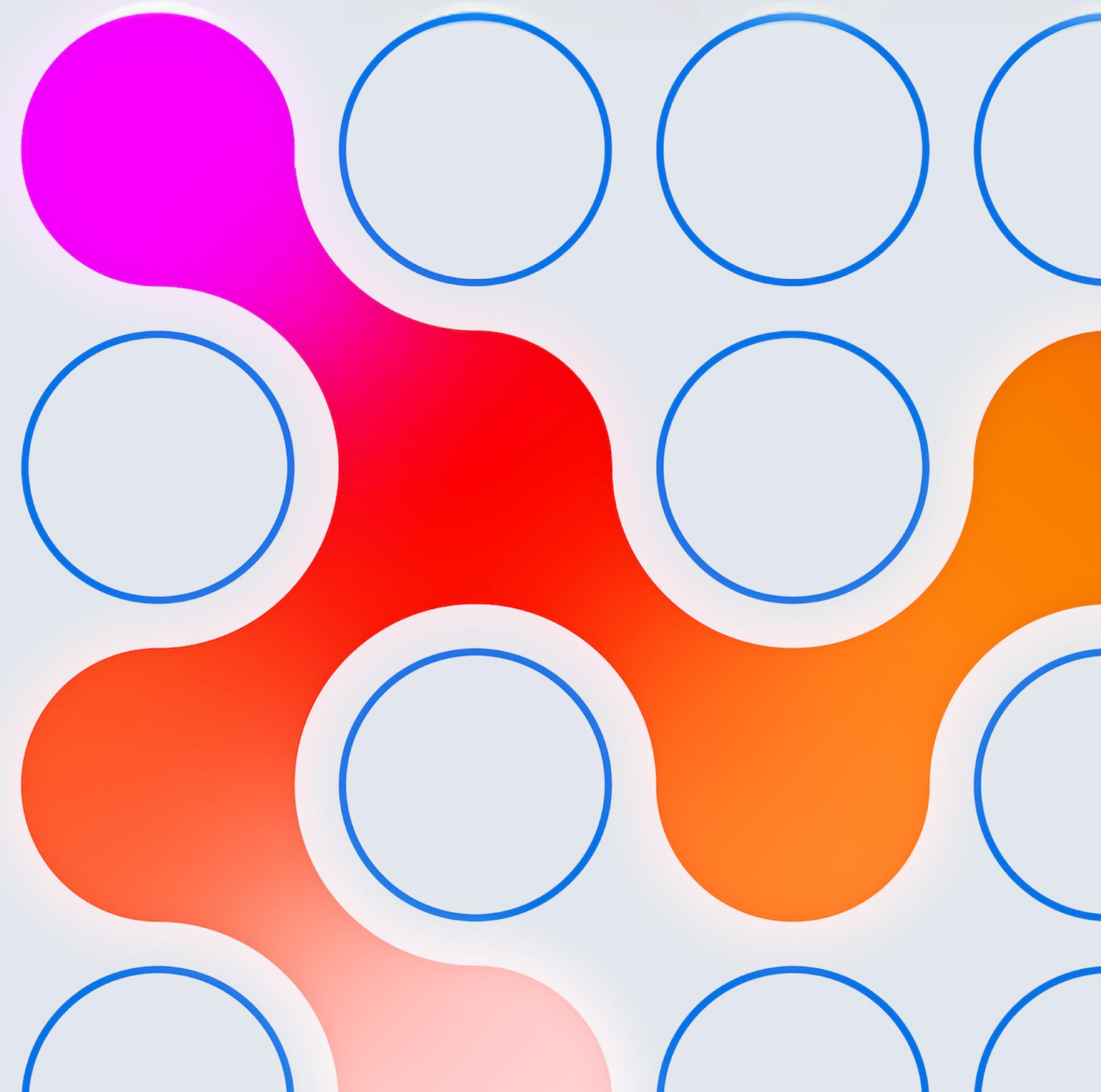
The key insight:

Each era removed a constraint: Systems became more flexible, then scalable, then able to access language and finally able to reason. The convergence of all these advances was the requirement for agentic AI.

We have seen a similar pattern with software



Large language models (LLMs)



What LLMs actually do

LLMs are key components in agentic architectures.

Understanding what they can (and can't) do is fundamental in agentic systems design.

Capabilities

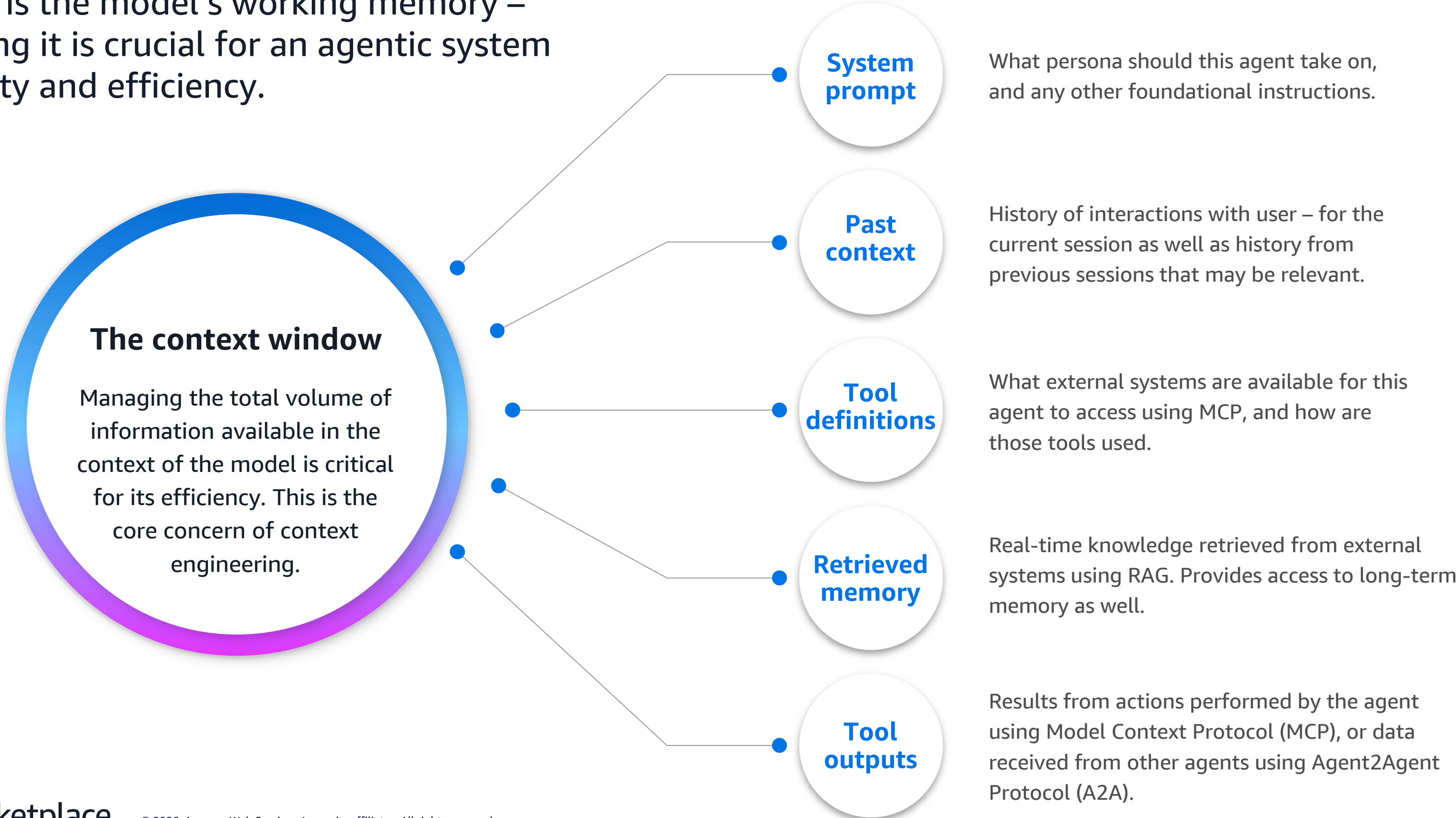
- Interpret natural language, can understand goals and intent
- Build a chain-of-thought, can reason
- Generate structured output
- Evaluate options and make decisions
- Generalize across domains without additional training or fine-tuning
- Process input using different modalities (text, images, audio)

Constraints

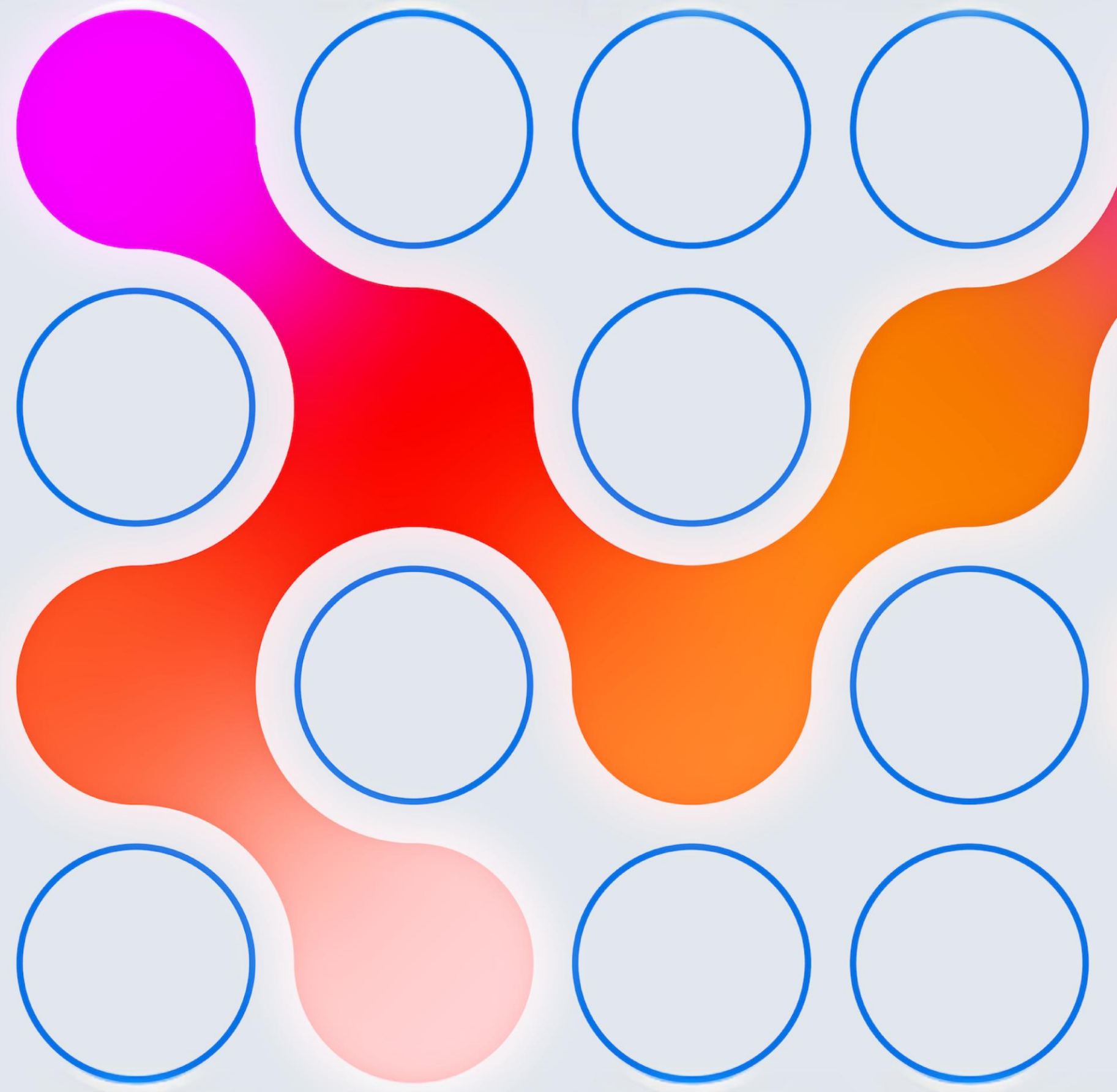
- Fixed context-window: working memory is limited
- No memory persistence
- Probabilistic and able to hallucinate facts confidently
- Latency and cost scale with context size
- Freshness of knowledge determined by training dataset
- Careful prompt and context engineering determine reliability

Context engineering and memory

Context is the model’s working memory – managing it is crucial for an agentic system scalability and efficiency.



From responses to actions



Two fundamentally different patterns

Basic LLMs: Request-Response

User input:

"Summarize this document"



LLM inference:

Single pass: one prompt, one output.



Response generated:

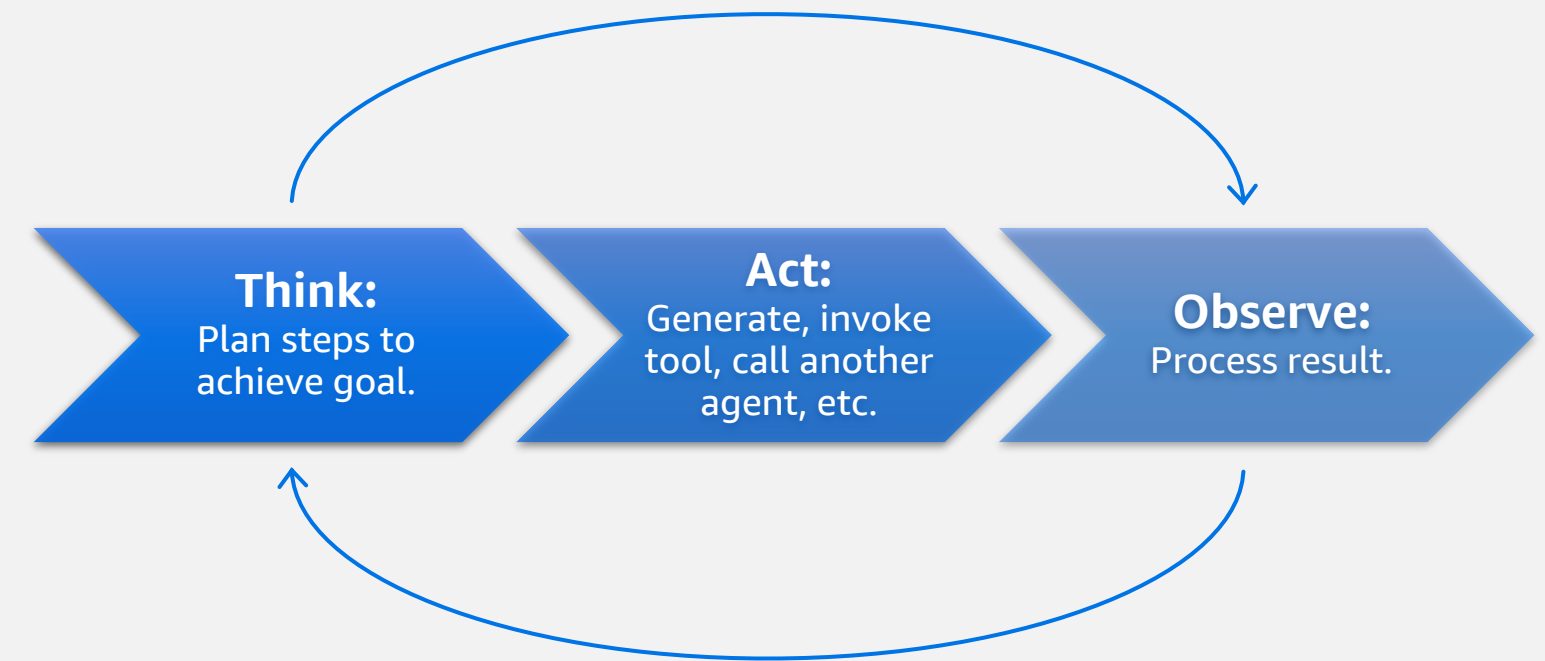
Text output as requested by the user or request for additional data.

Human must guide every step.

Agent: Think-Act-Observe Loop

User input:

"Scan my inbox for urgent tasks, add them to my To Do list, and email the requester with an ETA."



Loop until goal is achieved, deferred, or escalated.

LLMs + tools + data: Combined for agentic capability

Tools and external data sources enable agents to know about and act on the world.
Data grounds agents in reality.

Data and retrieval tools

- Vector search / RAG (Bedrock Knowledge Bases)
- SQL / NoSQL database queries
- Document parsing and indexing
- Web search APIs

Execution and action tools

- REST / GraphQL API calls
- Code execution environments
- Shell / CLI commands
- File system operations

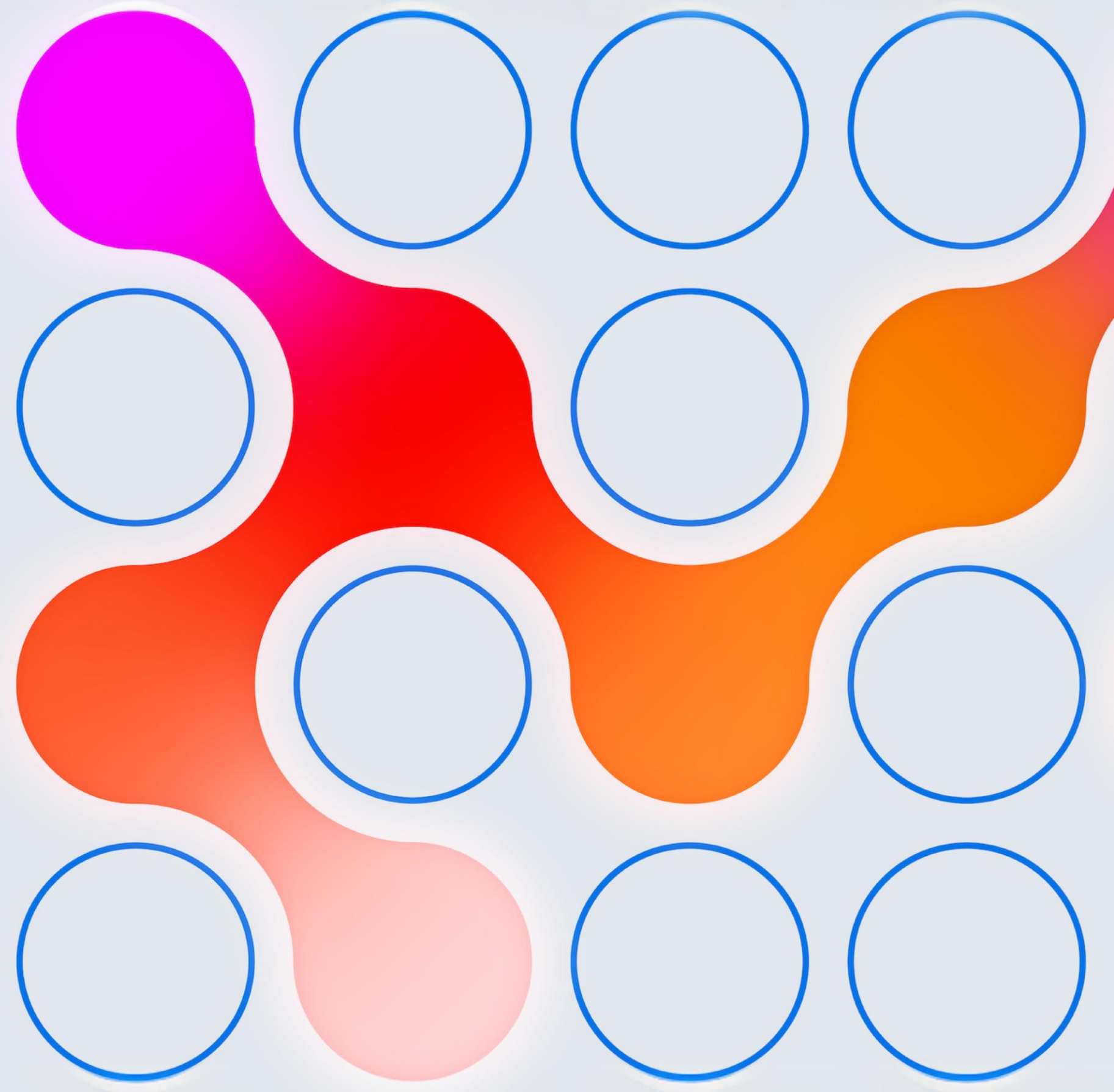
Communication tools

- Email and messaging services
- Slack / Teams integrations
- Webhook triggers
- Notification services

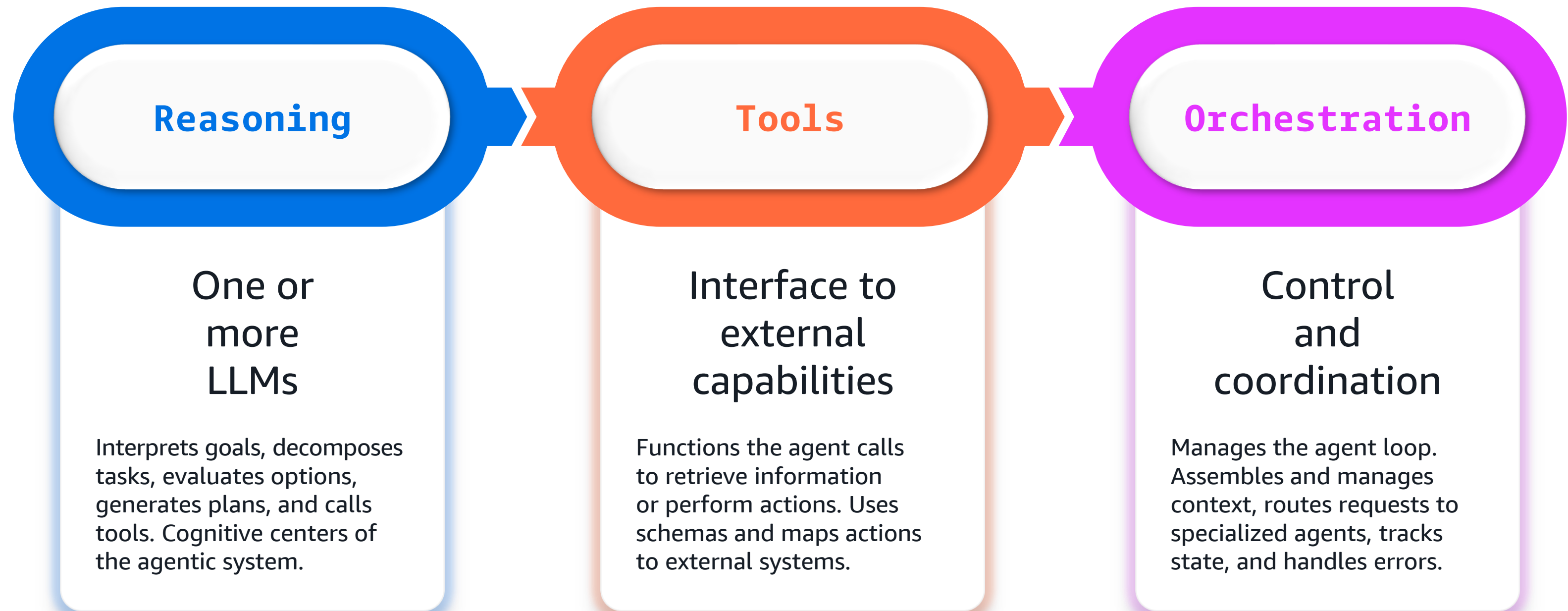
Orchestration tools

- Spawning sub-agents
- Step Functions workflow triggers
- Lambda invocations
- Human approval request

Agentic systems architecture



3 essential sets of capabilities compose every agentic system



Architectural design patterns

Single-agent loop

One agent, one loop, one goal. Simpler to build, debug, and maintain.

 **When boundaries are clear, toolset is limited and works in a single knowledge domain.**

Supervisor-Worker

A coordinator agent delegates tasks to specialized worker agents. Tasks are executed in sequence. Enables specialization and reduces complexity per agent.

 **When tasks require multi-domain knowledge and workflows are structured.**

Parallel/Swarm

Multiple agents share responsibility passing control directly to each other working in parallel and converging on a result.

 **When workflows are not structured or undefined, suitable for open-ended goals and when logic must be tested or redundancy is required.**

Human in the loop

Agent pauses at defined checkpoints for human review, approval, or correction. Essential for high-stakes workflows, good starting point in the process of gradual automation.

 **When actions are delicate or sensitive, or when compliance requires it. Good for early implementations.**

Interoperability: MCP and A2A

Standardized contracts that make agentic systems composable and maintainable.

MCP: Model Context Protocol

What it is:

A standardized protocol that defines how LLM hosts connect to tools, data sources, and external capabilities. Think of it as USB-C for AI — a universal connector.

Why it matters:

Without MCP, each agent needs custom integration code for every tool. With MCP, tools are discoverable and reusable across agents, frameworks, and models.

Status:

Open standard by Anthropic, adopted across Cursor, Claude, Zed, and growing ecosystem. AWS services are developing native MCP support.

A2A: Agent2Agent Protocol

What it is:

A protocol for agents to communicate, delegate tasks, and exchange results with other agents — even across different platforms and frameworks.

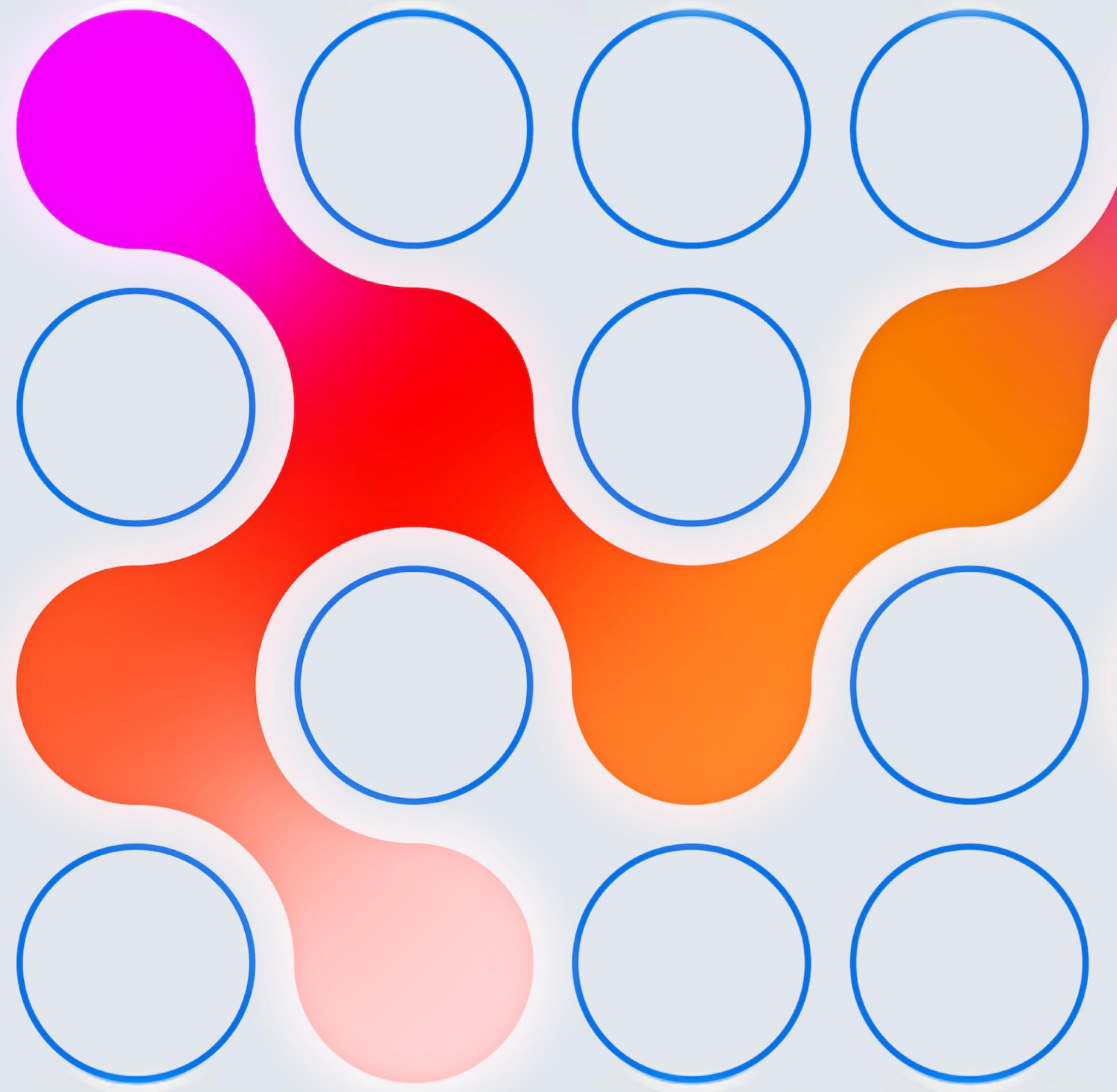
Why it matters:

Enables true multi-agent systems where agents from different vendors or teams can collaborate. Critical for enterprise architectures where not everything runs on one stack.

Status:

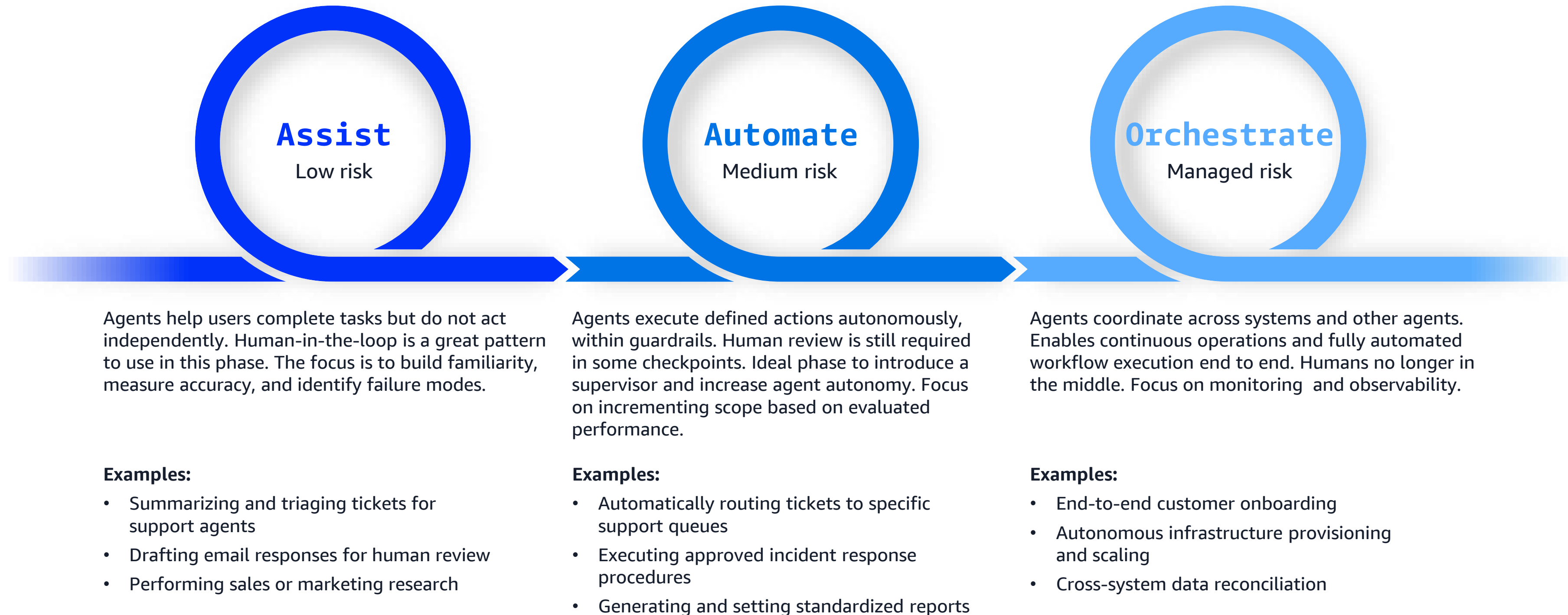
Emerging standards, gaining cross-vendor adoption. Defines task structure, messaging, authentication, and result formats for agent coordination.

How agents fit the enterprise



The journey to agent adoption

Build trust over time through measurements and iterative improvements



How to know if an agent works for a use case

Agents are not a universal solution to all problems. Selecting the right uses cases will define the overall success in adoption over time.

Agents are good for...

High variability

Tasks where input variations are numerous (ie. natural language) and achieving the objective cannot be codified in rules.

Cognitive load reduction

Tasks that deal with text or numbers and require working with large volumes of data.

Cross-system orchestration

Workflows that span across diverse APIs, databases or that require access to multiple knowledge domains.

Human capacity bottlenecks

Tasks that individually might be simple but occur at a volume that exceed the capacity of human teams.

Not so good for...

Deterministic workflows

Tasks where output must be exactly reproducible every time.

Fully structure input

If input is already perfectly structured, rules-based systems may be faster, cheaper, and more reliable.

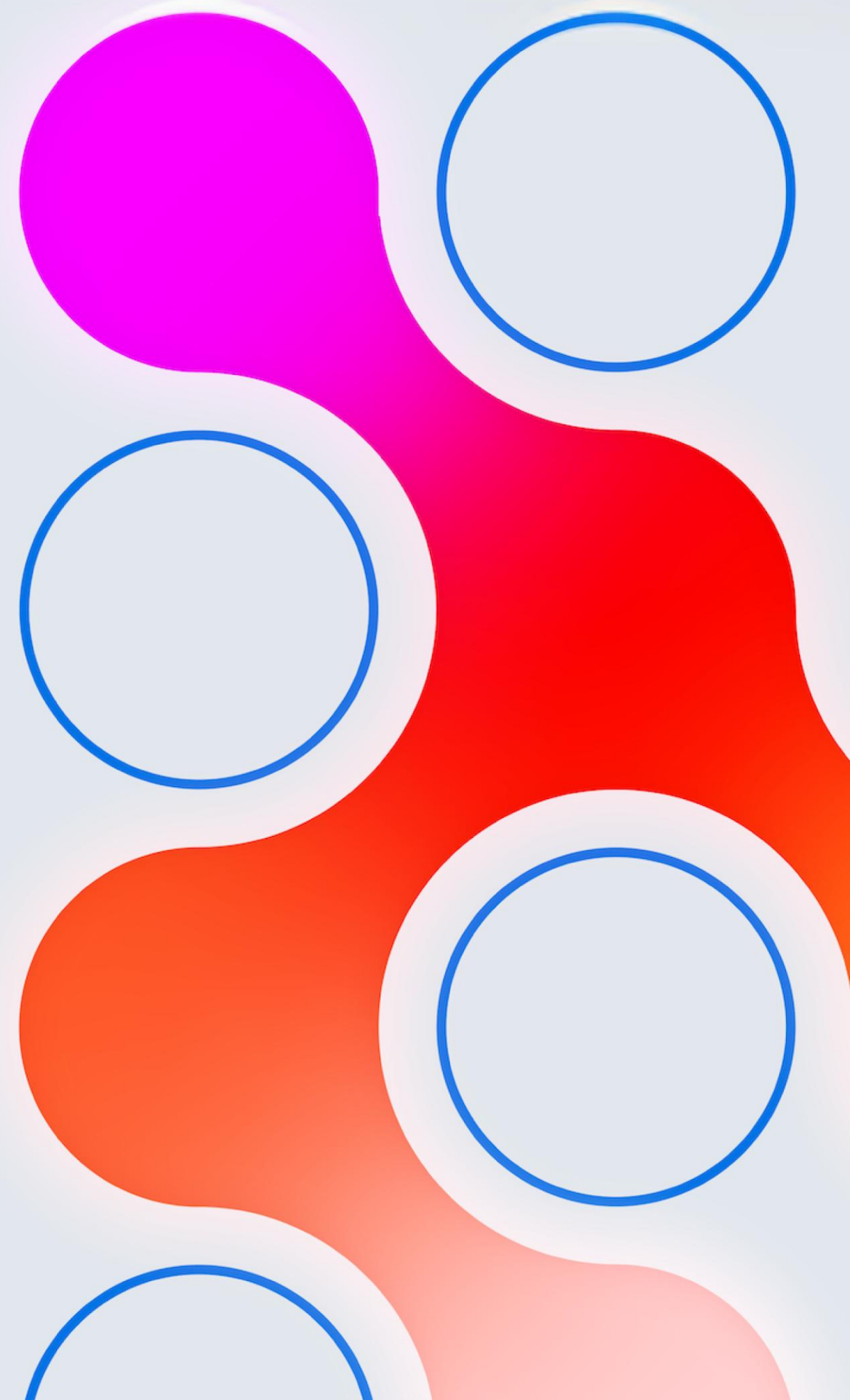
Zero-tolerance for error

Medical dosing, financial settlements, safety-critical systems – any use case where probabilistic output is unacceptable.

Simple, static automation

If a bash script or a simple DAG solves the problem, you do not need an agent.

Get started developing agentic systems



01

Design

Define scope, objectives, and constraints.

What is the agent's objective?

Clearly define the goals the agent will be qualified to take on. Be specific: Not 'help with customer service' but 'route incoming support tickets to the correct team based on category and urgency.'

What are the success criteria?

Define what 'done' looks like. How will you measure success? Routing accuracy? Time saved? Error rate? Without measurable criteria, evaluation is impossible.

What are the boundaries?

What can the agent do autonomously vs. what requires human approval? What systems can it access? What data is off-limits? Define this explicitly before building.

What is the minimal tool set?

Identify the absolute minimum tools needed to achieve the objective. Start smaller than you think you need. You can always add tools — but removing them later is harder when the system is in production.

What could go wrong?

Before building, enumerate failure modes. What happens if a tool returns an error? What if the LLM misinterprets the goal? How does the agent escalate? Design for failures, not just the happy path.

02

Develop

Cross-functional collaboration, software development lifecycle (SDLC), and iteration. All we have learned through software applies here, too.

Choose your frameworks and models

Just as with any other service, you don't want to do the undifferentiated heavy lifting. Select a framework that simplifies agent development by abstracting the low-level work required to scaffold or run agentic systems. Strands Agents, Amazon Bedrock and foundation models from Anthropic are great solutions to get started quickly without jeopardizing production readiness.

Build your development environment

Your engineers will need a reproducible and developer friendly environment to build, test, and iterate on agent development. Notice that any external tool and data source must be enabled for development in a secure way; for example, developers must have access to the dev instances of any MCP server their agent will rely on.

Establish SDLC and CI/CD

Versioning, continuous integration, delivery and deployment, immutable packaging, cross-environment promotion, and all the other capabilities we have learned and applied over the years to software development are just as applicable to agent development and agentic systems architecture.

03

Evaluate

Measure continuously. Probabilistic systems are about quality control.

Define quality criteria

Map evaluation dimensions to what you defined as success during design. Ensure that your system is instrumented to provide the necessary telemetry to objectively measure this criteria.

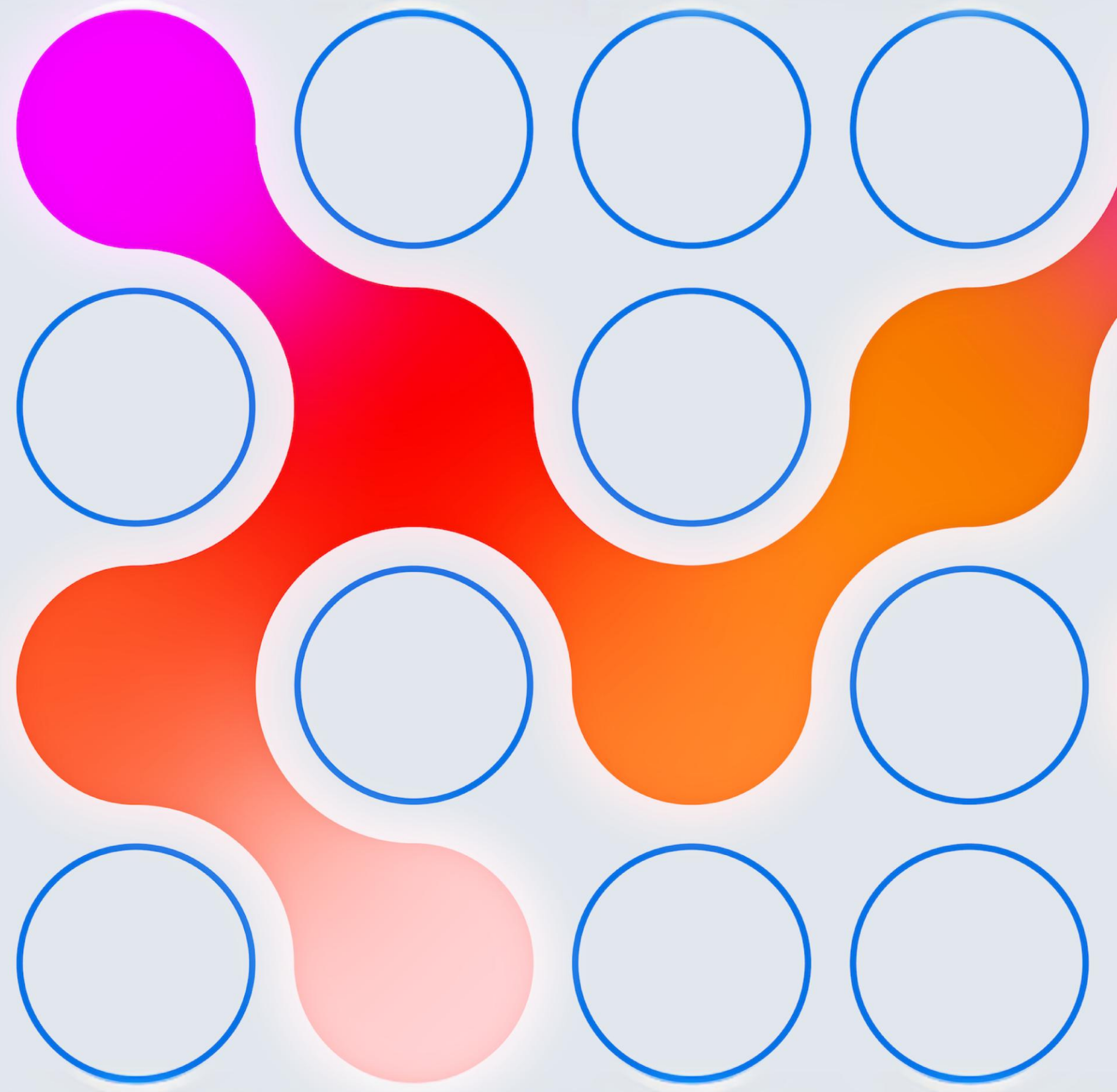
Use eval frameworks

You don't need to re-invent the wheel when evaluating the quality of your agents. Your chosen frameworks should provide readily available solutions, such as Amazon Bedrock AgentCore Evaluations, when using Strands Agents and Amazon Bedrock for your foundational stack.

Audit and monitor

Monitor responses to inputs based on correctness, faithfulness, conciseness, and response relevance. Monitor tool utilization in terms of selection accuracy, parameter accuracy, and error rates. There are many other primary metrics that can be used, dozens of them available off-the-shelf with Amazon Bedrock AgentCore Evaluations.

Real-world use case: **Agentic DevOps**



Building agentic systems in action

Step-by-step demonstrations for AI agent system development

Agentic DevOps

An agentic companion to automate and accelerate the usual workflows that DevOps and Platform Engineering teams handle every day.

- Inspect repositories and identify applications in them.
- Inspect the code and automatically identify infrastructure resources required by those applications.
- Produce infrastructure-as-code automatically for all infrastructure requirements.
- Automatically build CI/CD pipelines to test and build each identified application.
- Deploy the application to a development Kubernetes (K8s) environment and get the application running.
- Add Amazon CloudWatch alerts automatically for anything that triggers error logs.

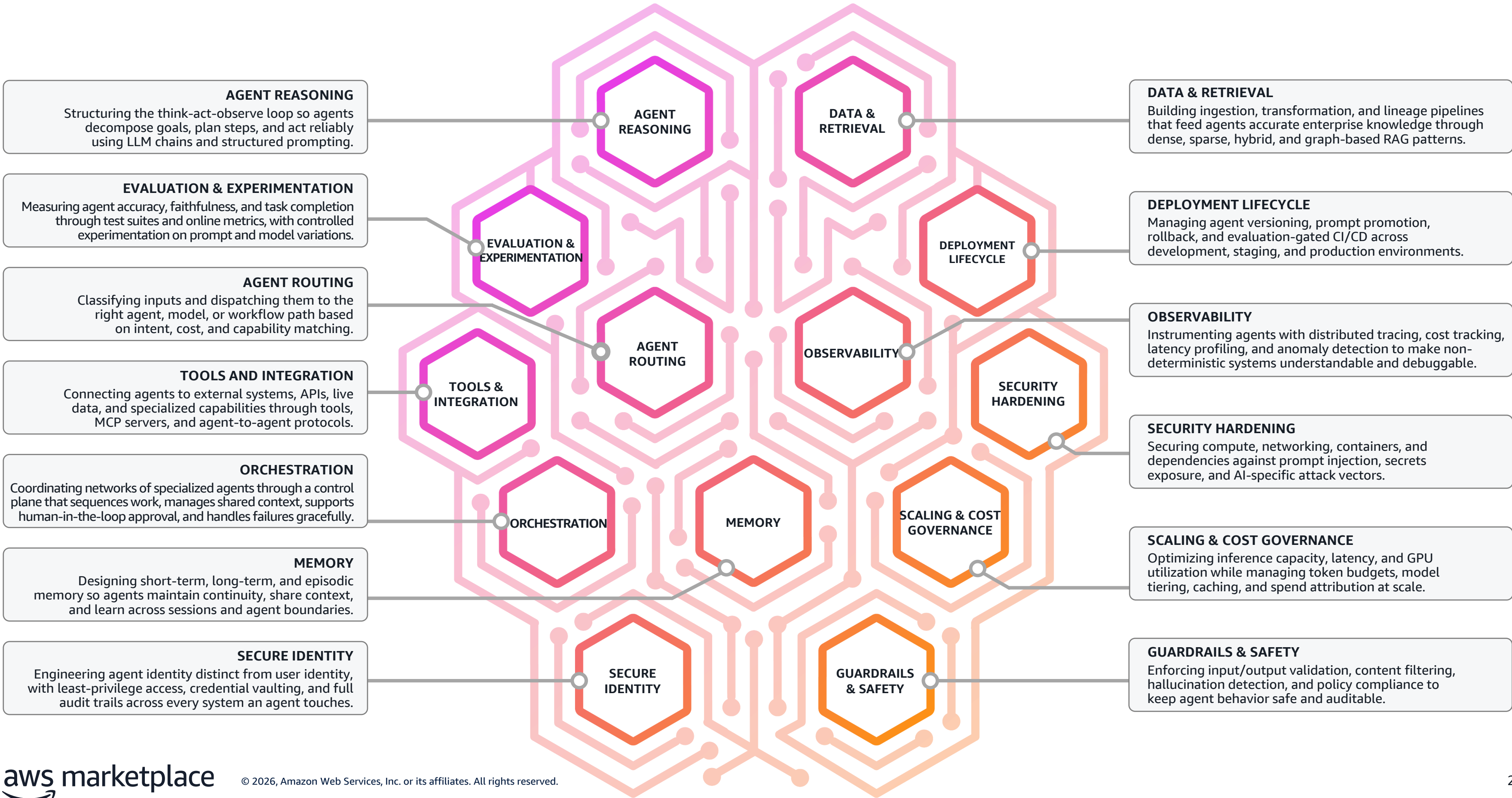
Each module in this series will extend the capabilities of the agentic system, by using the new concepts to enhance and increase the scope of what the agentic system and its agents are able to accomplish.

Not into DevOps?

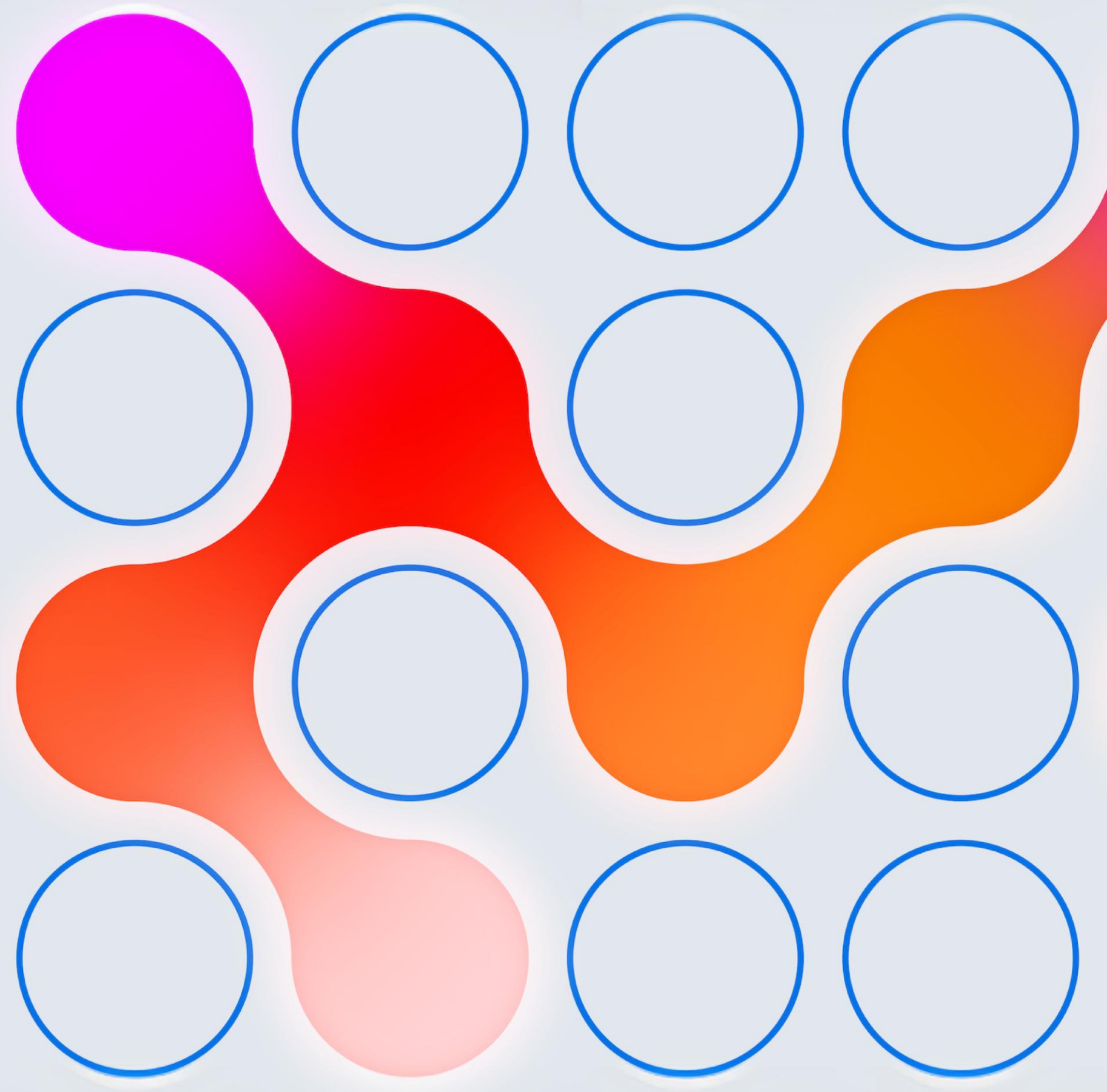
You can apply all the concepts demonstrated in the application to your own role.

The Agentic AI capability stack

13 capability domains to build production-ready agentic systems



Demo



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GUARDRAILS & SAFETY

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Guardrails AI

Arthur

Amazon Bedrock Guardrails

Amazon Bedrock AgentCore (Policy)

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Agentic AI tools and capabilities

To help you build, deploy, and operate agentic systems on AWS — available through your AWS account.

aws marketplace

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Why AWS Marketplace?

Streamline product discovery, procurement, and deployment.



AWS Marketplace offers pricing flexibility

Flexible consumption models

Ease into commitments

Free trials

Get started with a free trial before committing to purchase.



Bring your own license

Migrate to AWS with your existing product licenses. Intended for pre-existing customers.



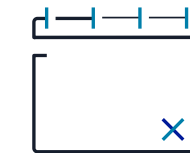
Usage-based subscriptions

Pay for software and compute capacity by the second or hour with no long-term commitments.



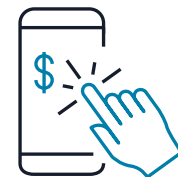
Contracts

Make a monthly, annual, or multi-year payment and receive a discount.



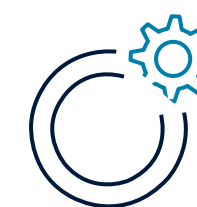
Private offers

Negotiate a custom price with a seller or Channel Partner. Accept the offer in AWS Marketplace.



Flexible payment scheduler

Set a custom billing frequency with your seller.



Next steps



Introduction to AI agents

Read a practical overview of AI agents, including how they relate to foundation models, access tools and data, automate workflows, and deliver measurable business value.



Building agentic systems on AWS

Explore the 14-module workshop series. Learn to design, orchestrate, and scale multi-agent systems – from foundational patterns to production deployment – with structured tutorials and vetted AWS Marketplace integrations.

Thank you!

